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Experimental investigation of the effect of IF-WS_2 as an additive in castor oil on tribological property

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ABSTRACT

The development of eco-friendly lubricant is the requirement of the times. In this paper, the modified IF-WS_2 nanoparticles are added to castor oil as additives to prepare mixed lubricant. The lubrication performances of the mixed oil are investigated by ball-on-disc rotary friction tester, and then the surface scratches are characterized by depth of field digital microscope, scanning electron microscope and Raman spectrometer. Experiments reveal that compared with the base oil, the friction coefficient and worn volume decrease by 27.4% and 47.2% after adding modified IF-WS_2 as additives under 300 rpm and 5 N conditions, respectively. From mechanism analysis, this improvement in lubrication performance is attributed to the polarity of castor oil and the unique pressure bearing structure and dispersion stability of modified IF-WS_2, which makes that the castor oil molecules can better adsorb around the modified IF-WS_2 to form heavy loading bearing friction oil film so as to reduce friction and wear. In summary, results indicate the mixture of modified IF-WS_2 nanoparticles and vegetable lubricating oil has an attractive prospect in the field of environmentally-friendly products.

1. Introduction

With the development of machinery manufacturing industry, friction and wear have become one of the main ways of energy consumption in the world, which accounts for about one third to one half [1]. Moreover, nearly 50% failure of mechanical assemblies originates from wear and friction. Therefore, how to reduce friction and wear better is becoming increasingly important. For traditional lubricating method, before the idea of solid nanoparticles as lubricant additives is prevalent, the lubrication between moving surfaces is usually achieved through synthetic or mineral base lubricants [2]. However, it is difficult for just mineral lubricants to meet the increasing lubrication needs of machinery and equipment, especially in the face of large construction machinery under the condition of heavy load lubrication. To tackle this problem, the idea of nanoparticles assisted lubrication is considered as a promising concept [3]. Based on this, scholars make a try to decrease the friction and wear on mechanical devices by using nanoparticles lubrication [4–6].

One of the most recognized and effective methods is nano additive lubrication [7]. Various kinds of additives, such as nano-oxides, nano--disulfide, nano-metallic elemental, and nano-silicates have been studied to reduce the friction and wear [8–11]. Since the discovery of fullerene-like tungsten disulfide (IF-WS_2) [12], it has showed an excellent tribological properties [13]. And due to its unique spherical structure and material properties, it has attracted a lot interests in researching different application areas of the IF-WS_2 [14–16]. In nano additives lubrication field, Joly-Pottuz et al. [17] used IF-WS_2 particles as additives in PAO base-stock lubricant to investigate the lubricating properties of mixed oil, results showed that the IF-WS_2 particles which were added to the PAO based oil could greatly decline the friction coefficient and abrasion loss. The lubricating properties of IF-WS_2 and IF-MoS_2 as mineral matrix lubricant additives were compared by Abate et al. [18]. From the experimental data, we knew that the IF-WS_2 seems to be better effective in friction reduction at the concentration of 1 wt%.

But on the other hand, although traditional mineral and synthetic lubricating oils have been extensively used to decrease friction and wear on mechanical parts. However, there are many shortcomings for them to meet the demand of environmental protection in lubricating field. There are some researchers indicate that about 50% of the lubricants around the world would end up coming into environment due to misuse, leakage or it's volatility [19,20]. To solve this problem, many scholars turn their eyes toward vegetable lubricating oils which lubricate better but easy to

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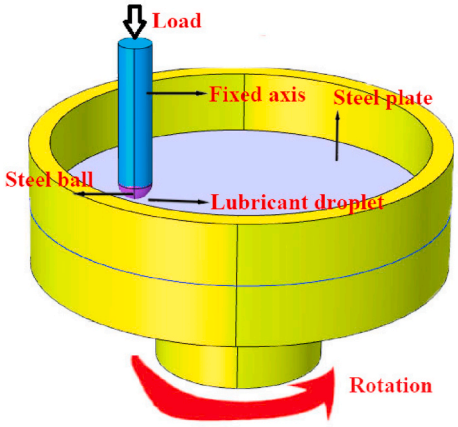

Fig. 1. The detailed schematic of the rotating tribometer.

be oxidized [20]. Many vegetable lubricating oils have great lubricity and excellent antifraying characteristics, compared with the petroleum-based oils [21]. Therefore, lots of scholars devote themselves to investigate the structure composition and lubrication mechanism of the vegetable lubricants [22,23] so that they can replace mineral oils in future. Moreover, just like the mineral oils, a number of scholars concern its performance as base oil of the additives lubrication [24–27]. Thereinto, the tribological properties of carbon nanotubes (CNTs) formed from oil fly ash were compared with that of the normal commercial carbon nanotubes through adding them to the sunflower oil (vegetable lubricant) by Salah et al. [28]. Results revealed that the tribological properties of the oil fly ash CNTs were better than the normal one's, and the oil fly ash CNTs as additives can greatly improve the lubrication performance of vegetable lubricants. And Wang et al. [29] modified carbon nanotubes by using nanocomposites, and the modified carbon nanotubes were served as nano-additives to rapeseed lubricating oil (vegetable lubricant), so as to elevate the friction reduction and anti-wear performance of the based oil greatly. Among many vegetable lubricants, castor oil has the better tribological properties and oxidation stability [30] by attributing to ricinic acid. Ricinic acid, the main part of castor oil, has a hydroxyl attached to its 12th carbon atom, owning more –OH groups than ordinary vegetable oils. Moreover, the percentages of linoleic acid and linolenic acid which belonged to the polyunsaturated fatty acids in castor oil are very small [31]. Besides, as a main research direction of additive lubrication, the improvement of dispersion by modification has been paid much attention by researchers [32–34].

Pioneering studies on the synthesis and modification of IF-WS_2 have continuously expanded the applications of IF-WS_2 as a lubricant additive [35–37]. Although many scholars have made in-depth researches on fullerene-like tungsten disulfide (IF-WS_2), especially on the nano additive lubrication field, most of them focus on using traditional petroleum-based oils as the base oils. However, as the concept of green environmental protection prevails, traditional mineral base-stock lubricants become more and more powerless in protecting environment and reducing pollution. On the contrary, vegetable lubricating base oils are becoming a prevailing trend, because they can show better lubrication performance and have less impact on the environment. In addition, the tribological properties and lubrication mechanism of IF-WS_2 modified by oleic acid as an additive of vegetable lubricants under heavy load conditions, such as large construction machinery, still need to be further explored.

In this study, based on the low speed and heavy load lubrication needs of large construction machinery, the parameters of this experiment (detailed in Section 2.2.2) are selected according to Hertz contact theory. Besides, in large construction machinery, there are many contacts of rotating friction, such as bearings, etc., so the rotating friction testing system, shown in Fig. 1, is selected. Based on the above equipment and parameters, we investigate the effect of modified IF-WS_2 as an additive on improving the tribological properties of castor oil through friction experiments and various characterization methods. Castor oil is used as the base oil and modified IF-WS_2 nanoparticles are dispersed into the base oil through stirring and ultrasonic. The lubricating performances of the mixed oil are tested by rotary friction tester. Subsequently, the wear scars are characterized by Raman spectrometer, scanning electron microscope (SEM) and depth of field digital microscope, etc. It can be found that the friction coefficient and worn volume of the mixture are greatly reduced, compared with the base-stock lubricant. These results verify that the addition of modified inorganic fullerene tungsten disulfide (IF-WS_2) nanoparticles greatly improve the friction-reduction and the anti-wear performance of castor oil. Moreover, 300 rpm - 5 N is found as the optimum working condition within the setting ranges for the lubricants exhibiting their best tribological performances. The significant improvement of the friction-reduction and anti-wear performances indicates that the combination of IF-WS_2 nanoparticles and castor oil has high potential research value at the field of tribology.

2. Experimental part

2.1. Preparation of materials and equipment

The castor oil, deionized water, anhydrous ethanol and the oleic acid used in the experiment were commercial available (Shanghai Aladdin Biochemical Technology Co., LTD). They are both analytical grade reagents. And the IF-WS_2 nanoparticles are prepared by the gas-solid reaction using WO_3 nanoparticles as initial material. The detailed preparation process was referred to Xu et al. [38]. Oleic acid was used for modifying and dispersing. According to mass ratio 1:5 (IF-WS_2: oleic acid = 1:5), IF-WS_2 and oleic acid were weighed to mix, and they were then dispersed by ultrasonic ball milling and reacted at 70 °C for 1h. After cooling, the products were cleaned and filtered with anhydrous ethanol several times, then washed and filtered with deionized water to neutral, and finally dried in a vacuum at 40 °C to obtain the modified IF-WS_2 as reserve. Five different concentrations' mixture of castor oil and modified IF-WS_2 (0.02 wt%, 0.05 wt%, 0.1 wt%, 0.2 wt%, 0.5 wt%) were prepared. In addition, homogenizer and ultrasonic cleaner were used to stir and disperse the mixed oil. As for the steel balls (Ø3 mm) and the steel disks (Ø48 mm, thickness of 1 mm) which were made of standard SUS304 stainless steel (hardness of 180–200 HV), their essential attributes were consistent with SUS304 stainless steel, but the dimensions of them were custom-made so as to ensure they can be used on the rotating tribometer. The reason why we choose SUS304 stainless steel is because the steel-steel contact is the most common in construction machinery which is served at low speed and heavy load condition, and SUS304 stainless steel is widely used in steel and is a representative one. Besides, stainless steel is used to protect the surface to prevent rusting. The tests were mainly performed via the MS-T3001 rotating tribometer, the detailed schematic of the MS-T3001 rotating tribometer was exhibited in Fig. 1. The steel ball contacts with the steel sheet through normal load, and the rotating base drives the steel sheet to rotate to produce rotating sliding friction with the steel ball, so as to simulate the actual movement of pin shaft.

2.2. Aggregation and tribology test

2.2.1. Aggregation test of mixed lubricating oil

In order to maintain the best lubrication performance, nanoparticle additives have to evenly scatter on the base-stock lubricant, and maintain dispersing stability without subsiding and aggregating in a long time. Therefore, the castor oil containing 0.2 wt% IF-WS_2 and 0.2 wt% IF-WS_2 modified by oleic acid (OA/IF-WS_2) were placed in different standing bottles to observe their dispersion. Besides, optical microscopy and ultraviolet visible spectrophotometer were also used to investigate

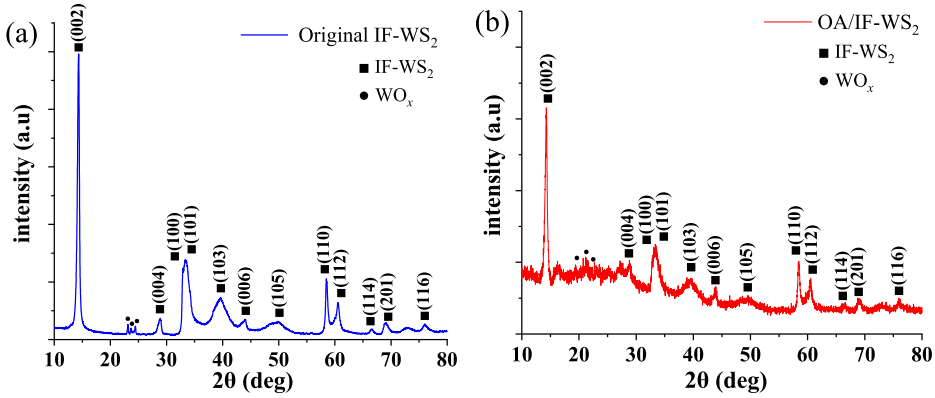

Fig. 2. X-ray diffraction patterns of IF-WS_2 before and after modification by oleic acid. (a) original IF-WS_2; (b) OA/IF-WS_2.

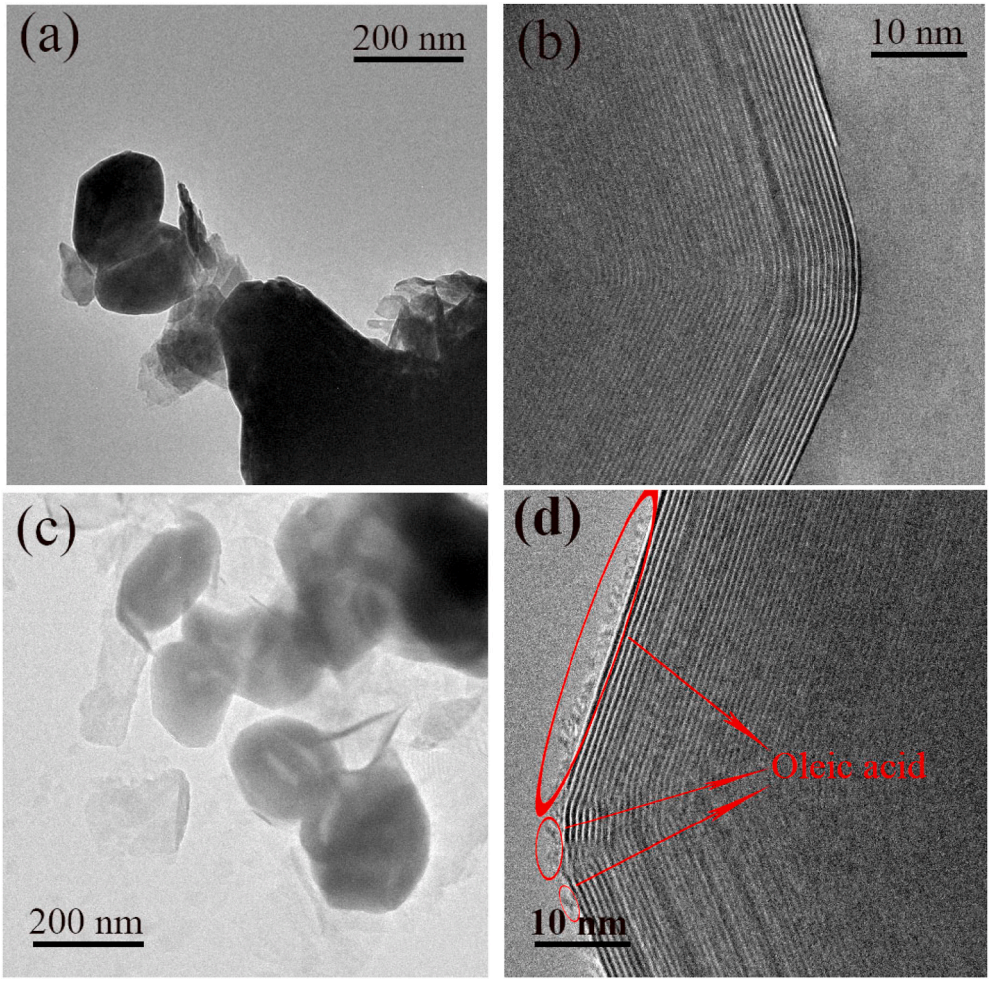

Fig. 3. TEM images of IF-WS_2 before and after modification by oleic acid. (a) low magnification image of original IF-WS_2; (b) high magnification image of original IF-WS_2; (c) low magnification image of OA/IF-WS_2; (d) high magnification image of OA/IF-WS_2.

the dispersion property.

2.2.2. Tribology test of different experiment conditions

Different concentrations of the OA/IF-WS_2 additives were dispersed in the castor oil by stirring and ultrasonication, respectively. Friction and wear experiments were performed on MS-T3001 rotary tribometer (Lanzhou Huahui Instrument Technology Co., Ltd) under different experimental conditions for 30 min. In the initial conditions of 4 N and 250 rpm, variable control method was used to obtain the best point of concentration from mixed oils with 0.02 wt%, 0.05 wt%, 0.1 wt%, 0.2 wt% and 0.5 wt% modified IF-WS_2. On this basis, the mixed oil, confirmed as the optimal concentration of nanoparticles, was applied to test the tribological properties at various revolving speeds from 50 r/min (0.0209 m/s) to 500 r/min (0.2093 m/s) and different applied loads from 3 N to 10 N, thus the optimum speed and load were determined. All the experiments were carried out under ambient temperature at around 20 °C–25 °C. Besides, all the steel balls and plates have been cleaned by ultrasonic cleaner before test to eliminate the influence of the

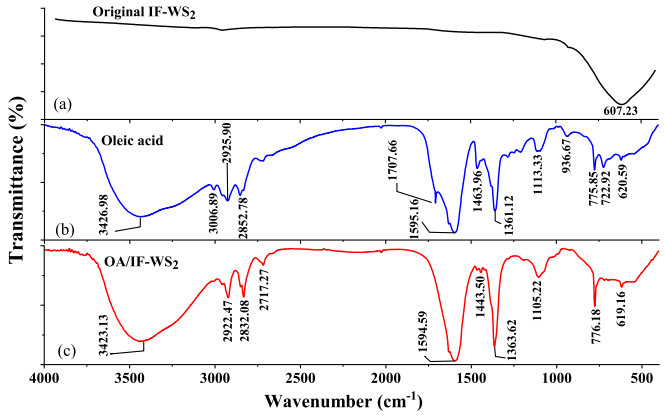

Fig. 4. FTIR images of the nanoparticles and modifier. (a) original IF-WS_2; (b) oleic acid; (c) OA/IF-WS_2.

contingency factor. The friction coefficient data during the test will be automatically recorded and stored by the friction sensor. In addition, each test was conducted three times and then the mean value was taken for record, so as to reduce the effect of accidentalia and ensure the reliability of the data.

2.2.3. Calculation of the maximum Hertz contact pressure

Usually, the maximum Hertz contact pressure is adopted to estimate the load is heavy or light. Due to the contact is between a sphere and a flat, the maximum Hertz contact pressure could be calculated using the formulas:

$$a^3 = \frac{3}{4} R_0 \left(\frac{1-\mu_1^2}{E_1} + \frac{1-\mu_2^2}{E_2}\right) P \tag{1}$$

$$\delta^3 = \frac{9}{16} \frac{1}{R_0} \left(\frac{1-\mu_1^2}{E_1} + \frac{1-\mu_2^2}{E_2}\right)^2 P^2 \tag{2}$$

$$q_0^3 = \frac{6}{\pi^3} \frac{1}{R_0^2} \frac{P}{\left(\frac{1-\mu_1^2}{E_1} + \frac{1-\mu_2^2}{E_2}\right)^2} \tag{3}$$

where a and R_0 are the radius of the contact circle and the steel ball, respectively, μ_1, μ_2 and E_1, E_2 mean the Poisson's ratio and Young's modulus of the steel ball and the steel plate. And δ is the change value in distance from the central point of the steel sphere to the plane of the steel sheets caused by elastic deformation, P represents the normal load, q_0 represents the maximum Hertz contact pressure. The range of applied load during the tests is from 3 N to 10 N, corresponding to the maximum Hertz contact pressure from 1.43 GPa to 2.14 GPa. In general, it will be recognized as the heavy load when the maximum Hertz contact pressure is more than 1.5 GPa. Therefore, the load used in the experiments is in the heavy-load region.

2.3. Characterization methods

Scanning electron microscope and depth of field digital microscope were applied to observe the worn appearance of wear scars. At the same time, the depth of field digital microscope was also used to show the worn profile and measure the wear volume of scratch on the steel sheet. Besides, Raman spectrometer (HORIBA Jobin Yvon LabRAM HR Evolution, France) was used to acquired the Raman spectrogram of worn surface, meanwhile, in order to make a better comparative analysis, the Raman spectrogram of IF-WS_2 samples was also determined.

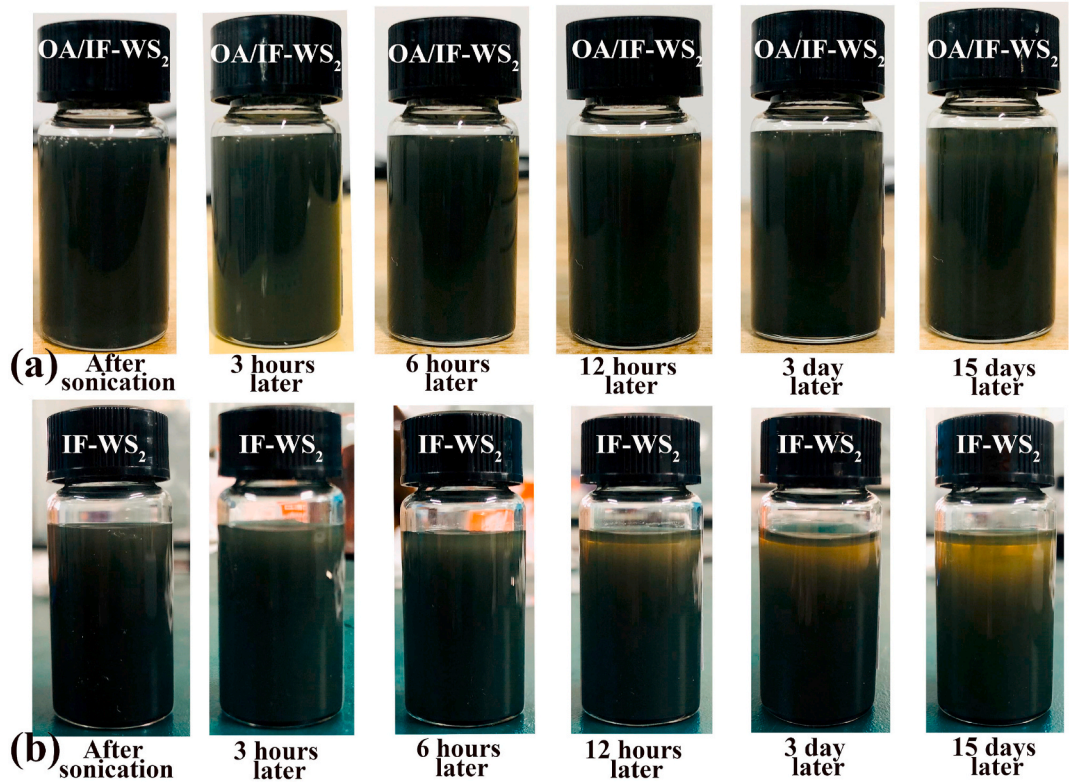

Fig. 5. Diagrams of dispersion stability over time. (a) dispersion property of 0.2 wt% OA/IF-WS_2 in castor oil, (b) dispersion property of 0.2 wt% original IF-WS_2 in castor oil.

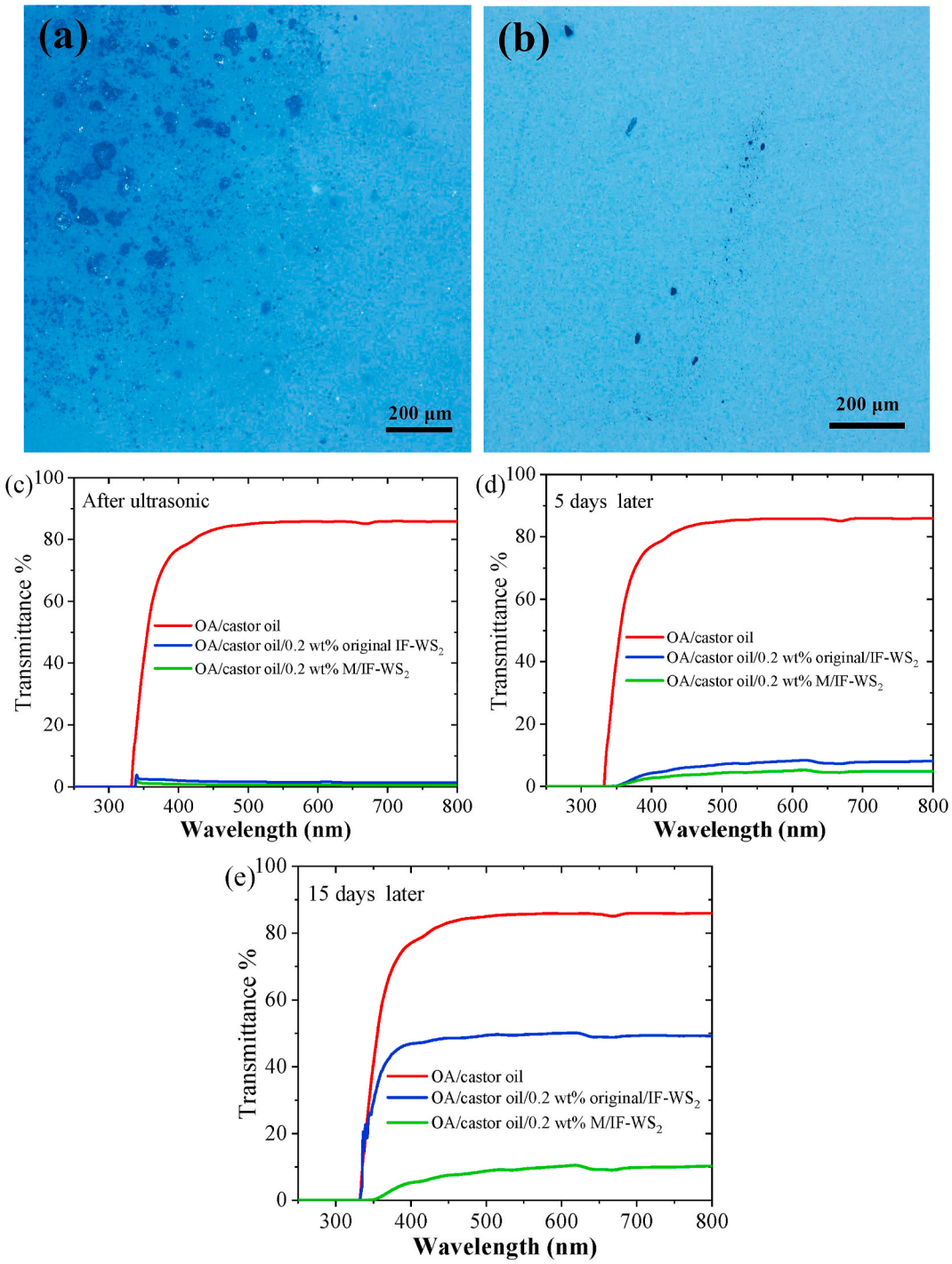

Fig. 6. Optical microscope images for different materials (M/IF-WS_2 and original IF-WS_2). (a) M/IF-WS_2; (b) IF-WS_2. UV-VIS transmittance results after standing for different time. (c) After ultrasonic; (d) 5 days later; (e) 15 days later.

3. Results and discussions

3.1. Characterization of nanoparticles and compatibility analysis of ester oil and nanoparticles

For better understanding the effect of oleic acid modification on the properties of IF-WS_2 nanoparticles, the IF-WS_2 nanoparticles before and after modification (original IF-WS_2 and OA/IF-WS_2) are characterized by XRD, TEM and FTIR.

3.1.1. X-ray diffraction analysis of the IF-WS_2 nanoparticles before and after modification

Compared with the standard 2H-WS_2 X-ray diffraction PDF card (JCPDS 08–0237), the X-ray diffraction image of the original IF-WS_2 showed in Fig. 2 (a) is actually quite similar to the standard card. However, there are subtle differences between them where the (002) peak of original IF-WS_2 slightly moves to the left and the (103) and (105) peaks widen somewhat [38]. The above analysis reveal that the nanoparticles used in this study have good crystallinity and different structure from that of 2H-WS_2. Besides, comparing the X-ray diffraction images of the IF-WS_2 before and after modification showed in Fig. 2 (a)

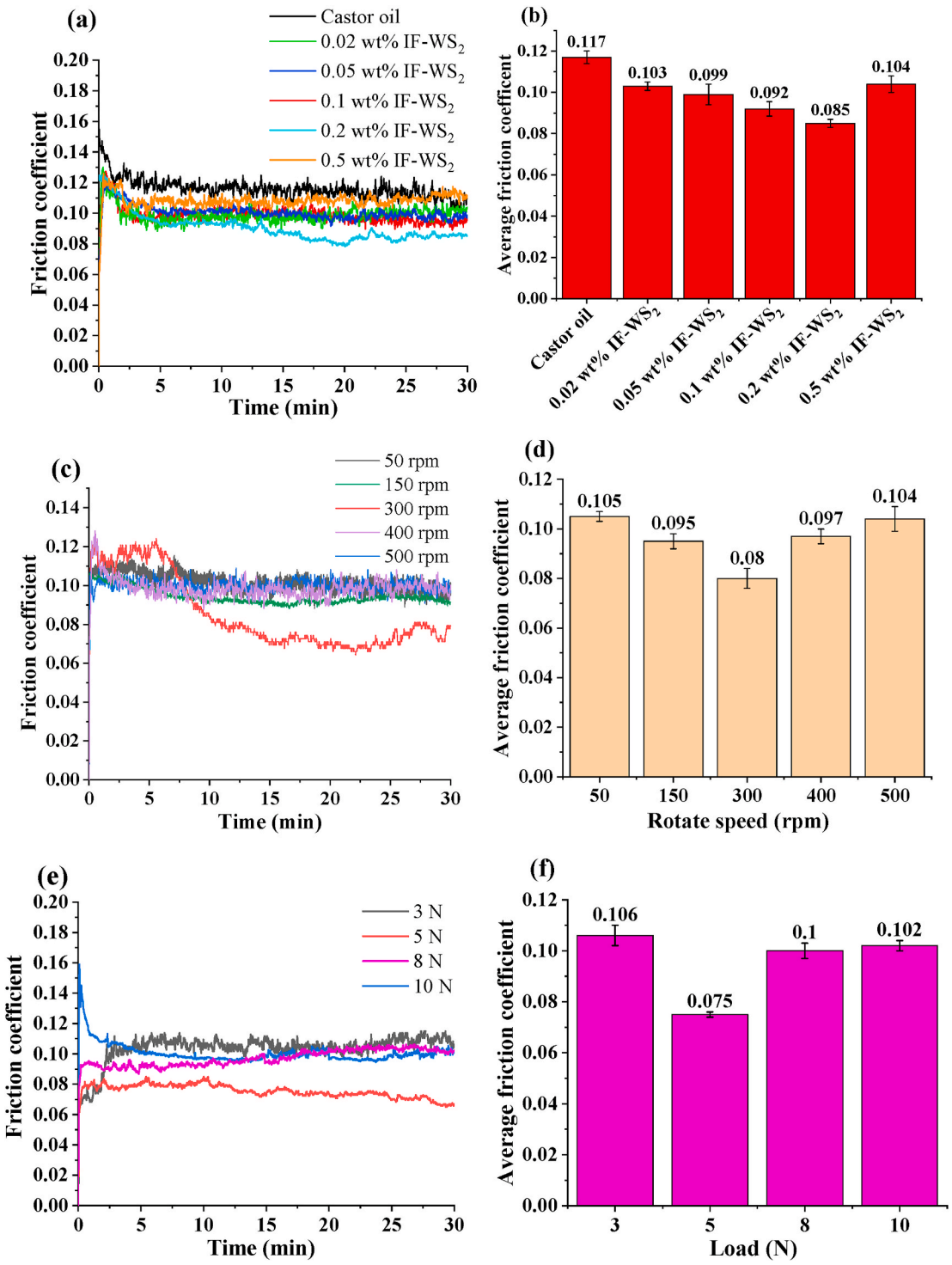

Fig. 7. Diagrams of the variation of friction coefficient along time and variation tendency under various experimental conditions. (a), (c) and (e) the transient variation of friction coefficient along time at various concentrations (250 rpm, 4 N), different rotate speeds (0.2 wt%, 4 N) and different loads (0.2 wt%, 300 rpm), respectively; (b), (d) and (f) trend diagrams of friction coefficient at different concentrations (250 rpm, 4 N), different rotation speeds (0.2 wt%, 4 N) and different loads (0.2 wt%, 300 rpm), respectively.

and (b), it can be found that the diffraction peaks of the OA/IF-WS_2 are shifted to higher angles and the spectral lines are broadened. According to Scherrer equation, the particle sizes of IF-WS_2 modified by oleic acid decrease.

3.1.2. TEM analysis of the IF-WS_2 nanoparticles before and after modification

Fig. 3 (a) and (c) show the low magnification TEM images of IF-WS_2 nanoparticles before and after modification. It is clear that the modified IF-WS_2 particles, with smaller aggregation and overlap, have disperse than those before modification. Moreover, according to the low magnification TEM images, the diameter of particles is roughly

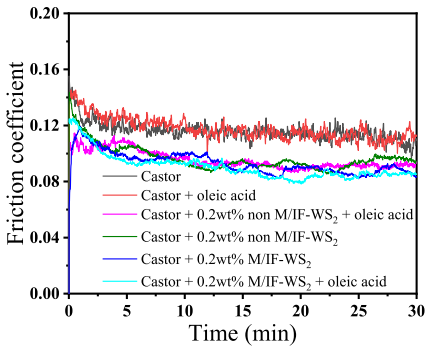

Fig. 8. Friction coefficient curves of different additives at the original condition (4 N-250 r/min).

distributed between 80 and 200 nm.

3.1.3. FTIR analysis of the IF-WS_2 nanoparticles before and after modification

Fig. 4 (a) shows the FTIR image of the original IF-WS_2. It can be found that there is only one peak at 607.23 cm^{-1} which is attributed to the W–S bond [39]. However, as can be seem in Fig. 4 (c), the FTIR spectra of the OA/IF-WS_2 has changed greatly comparing with that of the original IF-WS_2. Compared with oleic acid, the absorption peak of modified IF-WS_2 nanoparticles also changes. By comparing Fig. 4 (b) and (c), it can be found that the peaks belonging to methylene (–CH_2–) at 2910 cm^{-1} and 2850 cm^{-1} move towards the lower wave number. Furthermore, the absorption peak of carbonyl group at 1707.66 cm^{-1} disappears in the spectrum of the modified IF-WS_2 nanoparticles which suggests that the carbon-oxygen double bond may have broken and formed weak SH-WO-SH bond due to the coordination reaction between oleic acid and IF-WS_2 [40]. Besides, the other absorption peaks of the modified particles have changed relative to that of the oleic acid wave number, and the amount of movement exceed the normal error range of the FTIR spectrum [40]. Based on the above analysis, it is concluded that oleic acid is modified and coated with IF-WS_2 through chemical reaction rather than simple physical adsorption.

3.1.4. Dispersion stability analysis

Fig. 5 showed the dispersion property of IF-WS_2 and OA/IF-WS_2 particles as a function of time. As seen in Fig. 5 (b), the unmodified IF-WS_2 can only maintain dispersion stability for around 3 days. In contrast, the OA/IF-WS_2 showed better dispersion stability (shown in Fig. 5 (a)) which still keeps a evenly dispersion after 15 days.

Except for natural settlement experiments, optical micro-graphs of nanoparticles agglomeration sedimentation at low magnification and transmittance results from UV-VIS spectrophotometry (Fig. 6) were also employed to illustrate particle' s dispersion stability.

As is vividly showed in Fig. 6 (a), the modified IF-WS_2 with oleic acid show a significant agglomeration distribution which can be attributed to surface charge accumulation caused by the larger specific surface energy of nanoparticles. On the contrary, the reunion phenomenon has been

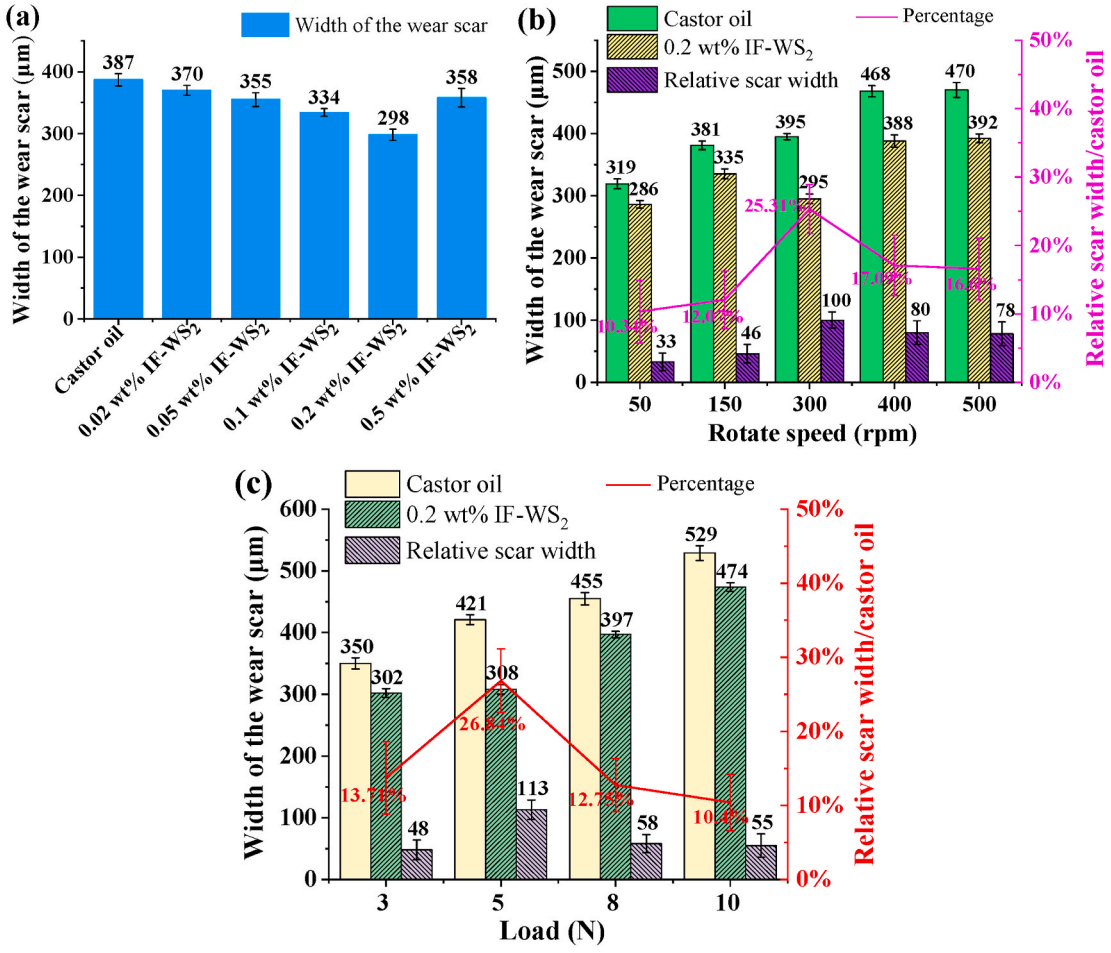

Fig. 9. Data of width of the wear scars under different conditions. (a) width of wear scar at different concentrations (4 N, 250 rpm); (b) width of wear scar and percentage improvement in wear reduction performance at different rotate speeds (0.2 wt%, 4 N); (c) width of wear scar and percentage improvement in wear reduction performance at different loads (0.2 wt%, 300 rpm).

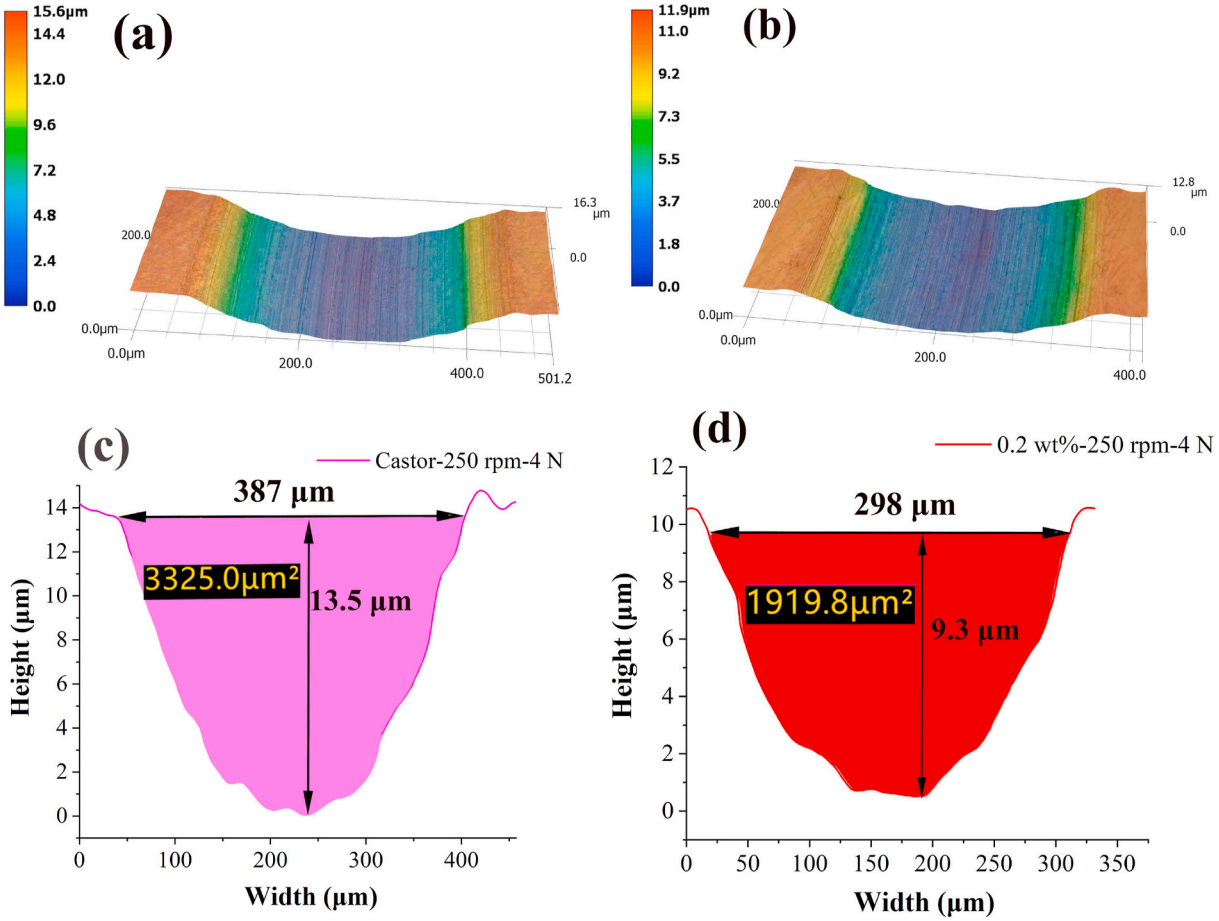

Fig. 10. Images about surface appearance and cross section profile of wear scars at different concentrations. (a) and (b) surface appearance of the wear scars at 0 wt % and 0.2 wt% (250 rpm, 4 N), respectively; (c) and (d) cross section profile of wear scars under 0 wt% and 0.2 wt% (250 rpm, 4 N), respectively.

obviously improved in Fig. 6 (b) when employed modified IF-WS_2 as nano-additives. This can be interpreted as that oleic acid suspended in the outer layer of IF-WS_2 through chemical modification has good compatibility with castor oil, thus increasing wettability of IF-WS_2 in Pao 6 and reducing its surface energy. Besides, it will produce new repulsive energy due to the presence of suspended oleic acid so as to weaken intermolecular attraction. Therefore, it can effectively prevent the contact between particles, reduce the agglomeration phenomenon and maintain a stable state. Besides, the dispersion stability of the nanoparticles is also measured quantitatively by the transmittance in the ultraviolet spectrophotometer.

It means that the system has the best dispersion stability if the transmittance of the system is the smallest. In Fig. 6 (c), after ultrasonic treatment, the transmittance of all the nano lubricants is less than 2%, which indicates that the nano lubricants are well dispersed. With the increase of settling time, the transmittance of unmodified nano lubricant increased to about 10% after 5 days (Fig. 6 (d)). 15 days later (Fig. 6 (e)), the transmittance of the unmodified nano lubricant is greatly increased to 50%, indicating that a large number of nanoparticles have settled. On the contrary, the transmittance of the modified nano lubricant is only about 10%, and the particle dispersion stability is excellent. All results indicate that the modified IF-WS_2 can greatly improve its dispersion stability in castor oil compared with that of unmodified IF-WS_2.

3.2. Influence of modified IF-WS_2 on friction-reduction and abrasion resistance

3.2.1. Friction-reduction performance

Fig. 7 (a) and (b) exhibit the evolution of friction coefficients with time at different concentrations (in the case of 250 rpm, 4 N) and the variation trend of the mean of the friction coefficients at various concentrations. In Fig. 7 (a) and (b), it is obvious that the corresponding average friction coefficient gradually decreases from 0.117 to 0.085 as the concentration of modified IF-WS_2 nanoparticles increases from 0 wt % to 0.2 wt%. However, when it surpasses 0.2 wt%, average friction coefficient increases with the increasing concentration of the modified IF-WS_2 nanoparticles. At the same conditions, the average friction coefficient of the mixed lubricant added with 0.2 wt% modified IF-WS_2 nanoparticles decreases by 27.4% compared with that of the castor oil.

These results suggest that 0.2 wt% nanoparticles provide the best lubrication effect. It is attributed to the excellent friction reduction and anti-abrasion characteristics of IF-WS_2 particles [13]. When the concentration is less than 0.2 wt%, the amount of modified IF-WS_2 particles involved in the friction surface will augment with the increasing concentration. However, when exceeds 0.2 wt%, the excess nanoparticles will subside and aggregate with the increase of the concentration, thus forming large particles to prevent the surface from sliding relative to each other.

As for Fig. 7 (c), it shows the transient variation of friction coefficient along time at 300 rpm and 500 rpm (0.2 wt%, 4 N), which reveals that applying medium rotate speed (300 rpm) is better for lubrication. Fig. 7 (d) shows that the variation tendency of average friction coefficient is increased after an initial decrease with the increase of rotate speeds. When at the condition of 0.2 wt%, 300 rpm and 4 N, the average friction coefficient reaches its minimum (0.08). Comparing with the case of 500 rpm, the minimum average friction coefficient has a decrease by around 23.1%.

The transient images about the friction coefficient under various

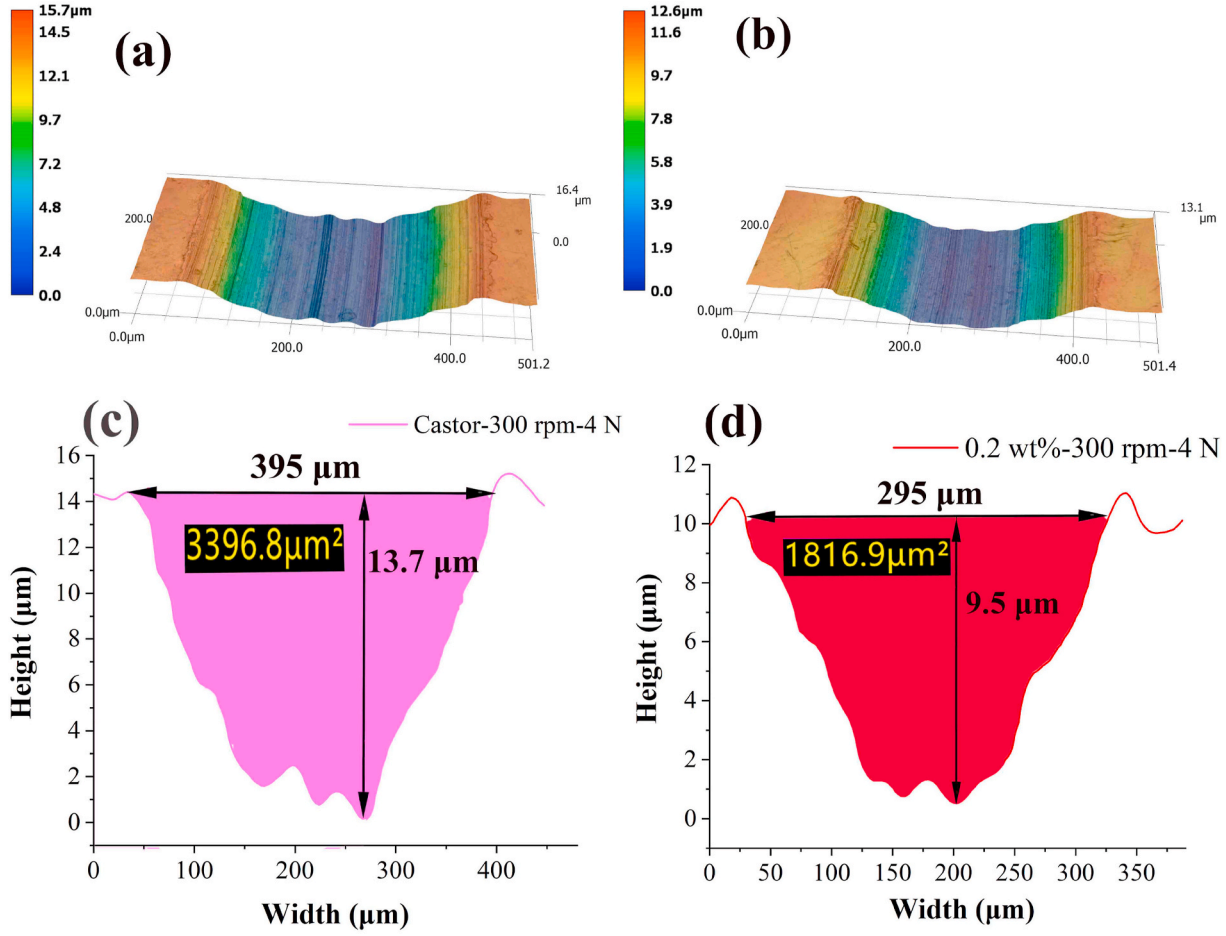

Fig. 11. Images about surface appearance and cross section profile of wear scars at optimal speed (300 rpm). (a) and (b) surface appearance of the wear scars at 0 wt % and 0.2 wt% (300 rpm, 4 N), respectively; (c) and (d) cross section profile of wear scars under 0 wt% and 0.2 wt% (300 rpm, 4 N), respectively.

loads (0.2 wt%, 300 rpm) and the variation tendency of the average friction coefficient at various applied loads are showed in Fig. 7 (e) and (f), respectively. Fig. 7 (e) indicates the better lubrication under 5 N compared with 10 N. Variation tendency of the average friction coefficient under different loads, exhibited at Fig. 7 (e), also decreases first and then increases, indicating that there is an optimal load (5 N) within the setting load ranges. In optimal load case, the friction coefficient (≈0.075) decreases by 26.5% compared with that in the case of 10 N.

From the above analysis, it can be found that there is an optimal speed and load to make the lubrication achieve best impact within a certain range. The possible causes are discussed in detail at Section 3.4.

In addition to the above analysis, in order to explore the influence of the oleic acid and the modified IF-WS_2 on friction-reduction properties, different friction coefficient curves of different additives at the original condition (4 N-250 r/min) are listed in Fig. 8. From Fig. 8, when only oleic acid is considered, the friction coefficient curves with and without oleic acid are basically the same, which reveals that there is few direct influence for oleic acid on friction-reduction properties, its impact on friction-reduction performance depends more on the suspension and dispersion of nanoparticles. However, from the friction coefficient curve of the M/IF-WS_2 (modified IF-WS_2) and non M/IF-WS_2 showed in Fig. 8, it can know that the friction-reduction property of the M/IF-WS_2 has a slight improvement (about 8%) compared with that of non M/IF-WS_2 when considering whether the IF-WS_2 were modified.

According to the above analysis, it can make a conclusion that whether oleic acid is added or not has little effect on the friction-reduction performance of lubricant, while the modification or otherwise of IF-WS_2 has a certain influence on the friction-reduction property.

3.2.2. *Anti-wear property*

As we can see, Fig. 9 (a) mainly exhibits the width of wear scars at different concentrations (250 rpm, 4 N). It is obvious that the variation trend of wear scar width at different concentrations is consistent with that of average friction coefficient under the same conditions. At 0.2 wt %, the width of worn scratch decreases by 23% compared with that of base oil, once again, revealing the best concentration (0.2 wt%) for lubrication.

It is attributed to the unique layered spherical structure and stable chemical and physical performances of IF-WS_2 particles, when the concentration of the modified IF-WS_2 does not exceed 0.2 wt%, it can act as tiny ball bearings and form a lubricating film during friction to prevent the direct contact of the solid surfaces, causing the greatly reduced wear. However, once the concentration exceeds 0.2 wt%, the excess of modified IF-WS_2 nanoparticles will cause agglomeration to form large particles which can prevent the relative motion between the moving parts, thus increasing wear loss.

Relative scar width (the difference value of the wear scar width lubricated by castor oil and mixed lubricant with 0.2 wt% IF-WS_2 under the same experimental conditions) is used to measure the lubrication effect. In Fig. 9 (b) and (c), it is found that the relative scar widths lubricated by 0.2 wt% modified IF-WS_2 are 100 μm and 113 μm at 300 rpm (4 N) and 5 N (300 rpm), and decrease by 25.31% and 26.84%, respectively, compared with that of castor oil. These data once again indicate the conclusion that there is an optimal rotate speed (300 rpm) and optimal load (5 N) within setting ranges to get the best lubrication performance. The detailed reasons are also analyzed in Section 3.4.

The cross-section profiles and the wear volume are measured

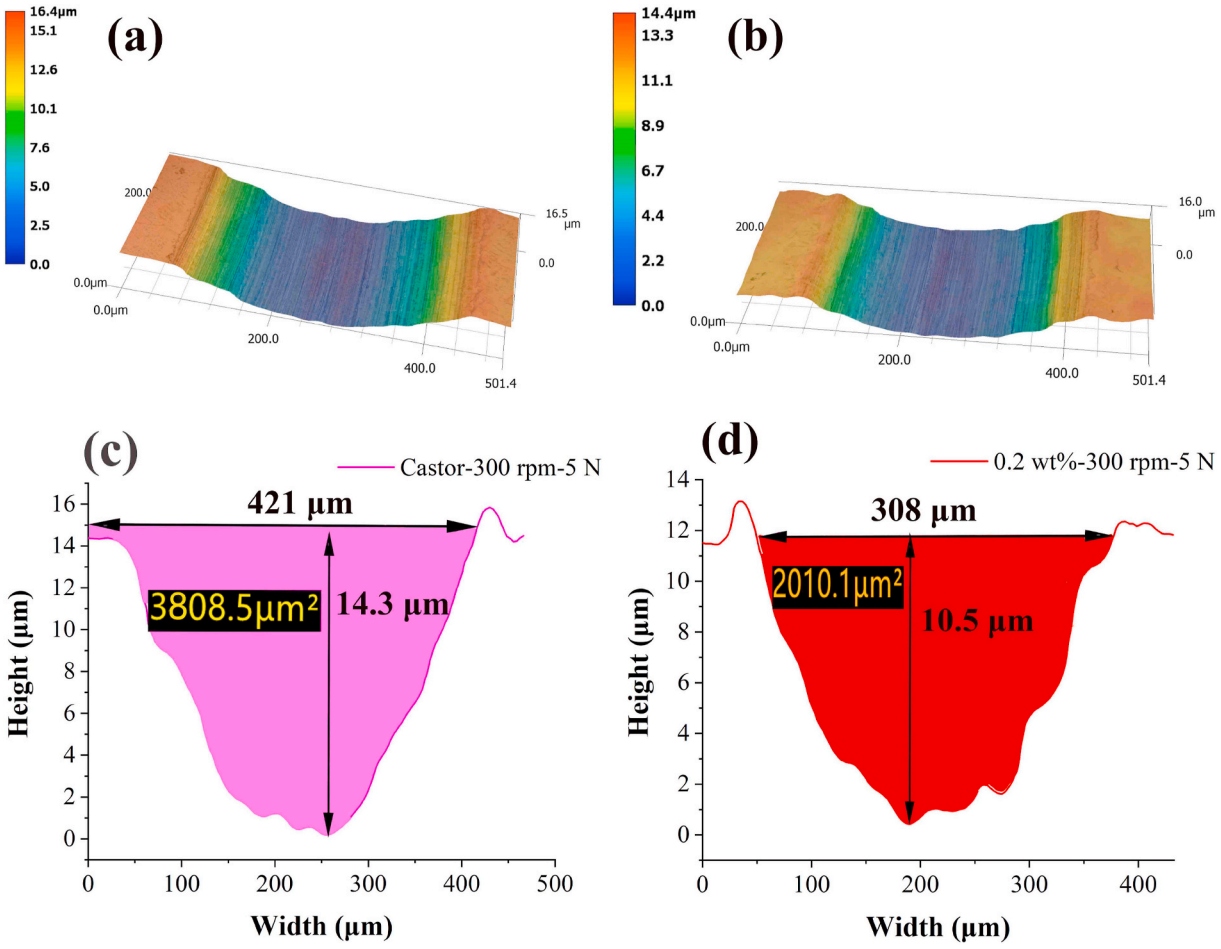

Fig. 12. Images about surface appearance and cross section profile of wear scars at optimal load (5 N). (a) and (b) surface appearance of the wear scars at 0 wt% and 0.2 wt% (300 rpm, 5 N), respectively; (c) and (d) cross section profile of wear scars under 0 wt% and 0.2 wt% (300 rpm, 5 N), respectively.

Table 1
Cross section parameters of wear scars at 0 wt%, 0.02 wt%, 0.05 wt%, 0.1 wt%, 0.2 wt%, 0.5 wt% cases (250 rpm, 4 N).

Parameter	250 rpm, 4 N					
	Castor	0.02 wt %	0.05 wt %	0.1 wt%	0.2 wt%	0.5 wt %
Width/μm (standard deviation)	387 (10)	370 (8)	355 (11)	334 (6)	298 (9)	358 (19)
Height/μm (standard deviation)	13.5 (0.6)	12.2 (0.3)	11 (0.4)	10.5 (0.3)	9.3 (0.2)	12.5 (0.4)
Volume/μm^2 (standard deviation)	3325 (153)	2688.1 (98)	2461.7 (85)	2114.8 (94)	1919.8 (63)	2521 (92)

Table 2
The wear volume lubricated by 0 wt% modified IF-WS_2 and 0.2 wt% modified IF-WS_2 under different speeds at 4 N.

Parameter	4 N				
	50 rpm	150 rpm	300 rpm	400 rpm	500 rpm
	castor, 0.2 wt%	castor, 0.2 wt%	castor, 0.2 wt%	castor, 0.2 wt%	castor, 0.2 wt%
Average volume (μm^2)	1410, 1144.2	2741, 1958.1	3396.8, 1816.9	3751.6, 2813.7	3915, 3613.7
Standard deviation	52, 46	91, 84	197, 35	156, 89	182, 97
Decrease percentage	18.85%	28.56%	46.51%	25.00%	7.70%

simultaneously by depth of field digital microscope, shown in Figs. 10–12 and Tables 1–3, respectively. Where, the cross-sectional area of the wear mark represents the amount of wear volume. It is worth noting that the wear data in Tables 1–3 is the average of three experimental measurement data.

The results under different concentrations at the initial conditions of 250 rpm and 4 N are showed at Fig. 10, limited by space, only the results of 0 wt% and 0.2 wt% concentrations are displayed. From Fig. 10 (a)–(d), it is obvious that lubricating by mixed lubricant with 0.2 wt% modified IF-WS_2, compared with base oil, not only the width of wear scar is greatly reduced (declined from 387 μm to 298 μm), but also the maximum wear scar depth has been significantly reduced (decreased from 13.5 μm to 9.3 μm). The corresponding worn volume also decreases from 3325.0 μm^2 to 1919.8 μm^2, therefore, the worn volume of the wear scar lubricated by mixed oil with 0.2 wt% modified IF-WS_2 particles has a decrease by 42.3% compared with the castor oil.

Table 1 show the widths, the wear heights and the wear volumes at different concentrations under the conditions of 250 rpm and 4 N. It is obvious that the wear volume data show a valley-shaped distribution, with 0.2 wt% as the lowest value (1919.8 μm^2) which is basically consistent with the trend of average friction coefficient and the width of the worn scratch under the same conditions. The consistent varying trends of these three parameters all prove the best positive role of 0.2 wt

Table 3
The wear volume of lubricated by 0 wt% modified IF-WS_2 and 0.2 wt% modified IF-WS_2 under different loads at 300 rpm.

Parameter	300 rpm			
	3 N	5 N	8 N	10 N
	castor, 0.2 wt%	castor, 0.2 wt%	castor, 0.2 wt%	castor, 0.2 wt%
Average volume (μm^2)	2370.2, 1639.4	3808.5, 2010.1	4203.7, 2904.2	4803.2, 3711.2
Standard deviation	86, 51	176, 96	162, 99	177, 152
Decrease Percentage	30.83%	47.22%	30.91%	22.73%

% modified IF-WS_2 nanoparticles in lubrication.

The appearance and cross section profile of the worn scratch at optimal speed (300 rpm) are shown in Fig. 11, other wear volume data are recorded in Table 2. In Fig. 11, it is obvious that at the optimal speed (300 rpm) and initial load (4 N), the wear volume decrease by 46.51% from 3396.8 μm^2 (in the case of 0 wt%) to 1816.9 μm^2 (in the case of 0.2 wt%). Compared with the initial conditions in Fig. 10 (d), the wear volume in Fig. 11 (d) should be increasing due to the increase of the rotate speed.

However, it has a slightly reduction. And the reduction in wear volume between 0 wt% and 0.2 wt% increases from 42.3% (initial conditions of 250 rpm) to 46.51% (300 rpm). It also proves optimum lubrication at 300 rpm case.

In Table 2, with rotate speed increasing, the relative wear volume (the difference of the wear volume between 0 wt% and 0.2 wt%) increases at the early stage and then decrease. It is clear that at 300 rpm, the relative wear volume reaches the maximum. The analyses of friction coefficient, width of worn scratch and worn loss under different rotate speeds prove that 300 rpm is the optimum speed for modified IF-WS_2 nanoparticles playing its role.

As for Fig. 12, similar to Fig. 11, it only shows the appearance and cross-section profile of the wear scars at optimal load (5 N). As can be seen in Fig. 12, at the optimal speed (300 rpm) and optimal load (5 N), the percentage of wear volume reduction is 47.22%, which has a slight increasing compared with that of the case at 300 rpm and 4 N (Fig. 11). Thus, indicating that the 5 N is the best load for lubrication once again. Moreover, the trend of the wear volumes in Table 3 also reveals that the anti-wear performance will be the best at 5 N.

In brief, all the analyses about the friction-reduction and anti-wear performance come to a conclusion that the mixture with 0.2 wt% modified IF-WS_2 particles exhibits the best lubrication performance under the 300 rpm and 5 N conditions.

At the same time, in order to illustrate the repeatability of data, the other two cross section profiles and wear volume of repeated experiments are shown in Fig. 13 and Table 4. At the same conditions, the similar results of repeated experiments ensure the reliability of statistical data.

Corresponding to Fig. 8, different cross section profiles of different additives at original condition (4 N-250 r/min) showed in Fig. 14 are mainly to illustrate the effects of adding oleic acid or not and IF-WS_2' s modification or not on anti-wear properties. By comparing Fig. 14 (a) (c) (e) with (b) (d) (f), it is obvious that the effect of the oleic acid on anti-wear properties is little if just considering the effect of adding oleic acid or not. While only taking into account the IF-WS_2' s modification or not, the modified IF-WS_2 nanoparticles (Fig. 14 (c) (d)) can slightly improve the anti-wear performance (about 10%) compared with that of original IF-WS_2 (Fig. 14 (e) (f)). These results, which are consistent with the friction-reduction, indicate that the improvement of friction-reduction and anti-wear properties has little to do with oleic acid, and has a positive correlation with the modification of IF-WS_2, but it is mainly due to the effect of nanoparticles themselves.

3.2.3. *Wear analysis of counterbodies (steel balls)*

In order to better evaluate the wear loss of tribosystem, a series of

Table 4
Wear volume of repeated experiments at different conditions.

Condition		250rpm-4N		300rpm-4N		300rpm-5N	
		0 wt%	0.2 wt %	0 wt%	0.2 wt %	0 wt%	0.2 wt %
Wear volume (μm^2)	01	3171.7	1857.3	3593.6	1852.0	3632.8	2106.3
	02	3478.3	1982.3	3200.0	1781.8	3984.2	1913.9
	03	3325.0	1919.8	3396.8	1816.9	3808.5	2010.1

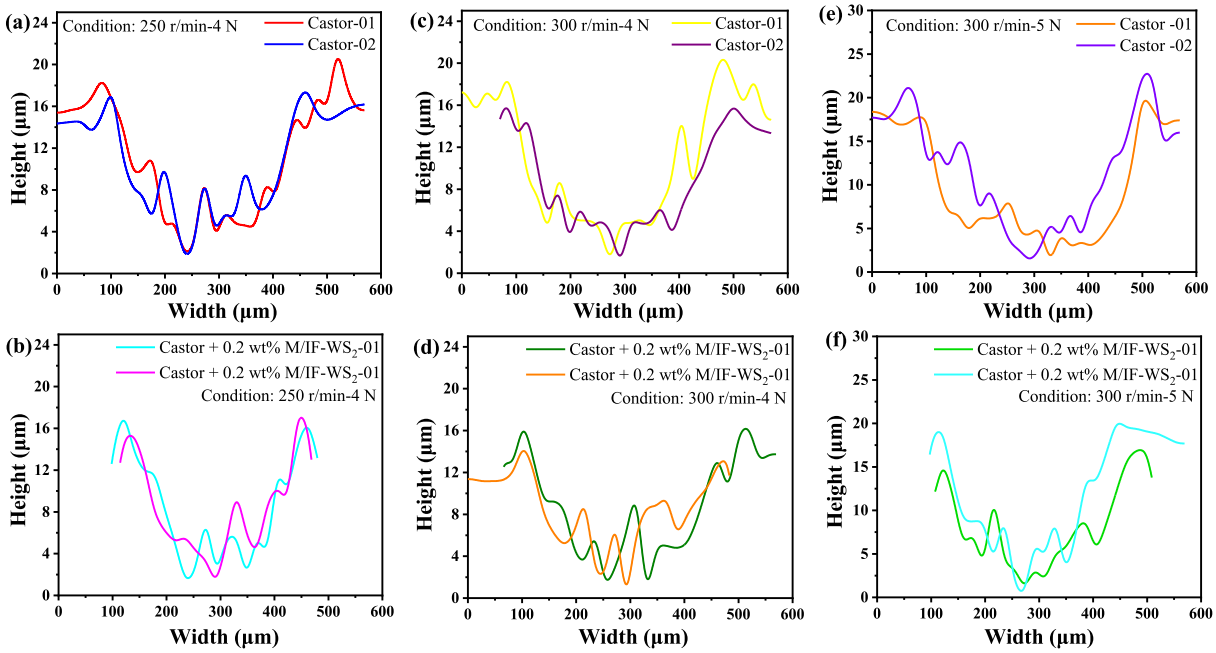

Fig. 13. The cross section profiles of repeated experiments at different conditions.

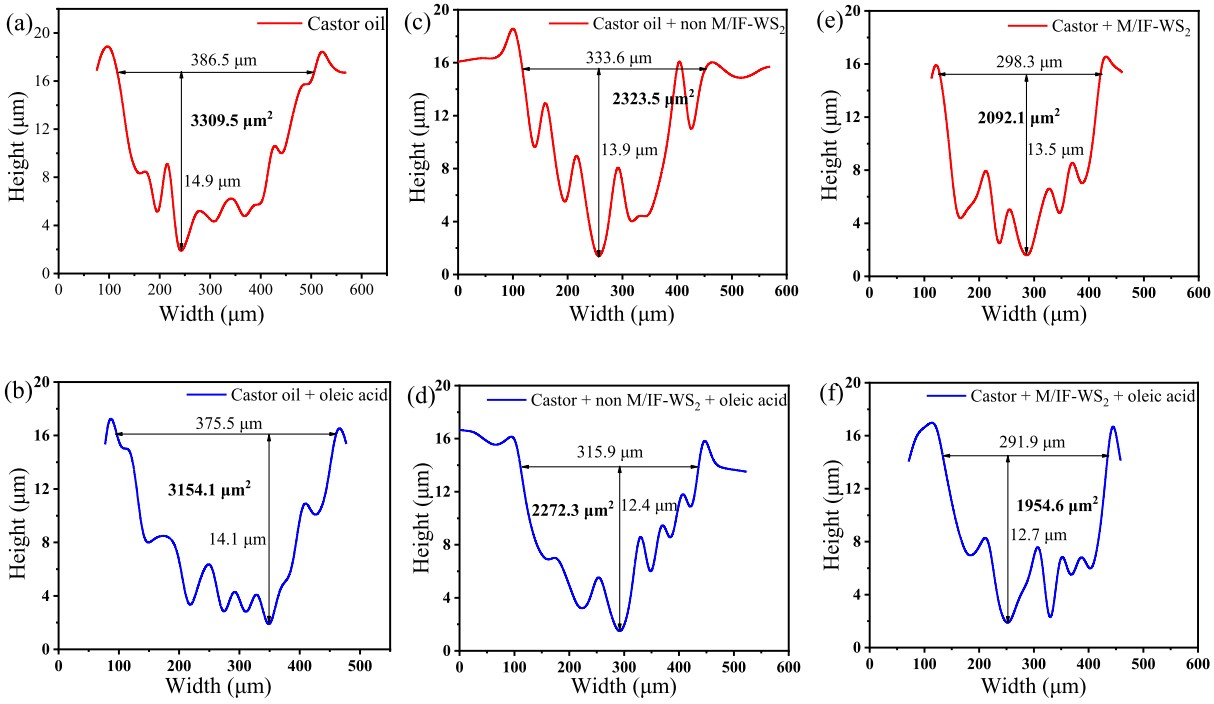

Fig. 14. Different cross section profiles of different additives at original condition (4 N-250 r/min).

correlation wear analyses are also carried out on the counterbodies (steel ball). Fig. 15 shows the microscopic morphology, three-dimensional image and section contour information of the wear marks of the steel balls obtained by the depth of field microscope.

According to the wear surface morphology shown in Fig. 15 a(I)-f(I), it can be clearly observed that the lubricating oil containing 0.2 wt% modified IF-WS_2 as the additives has significantly smoother wear marks, and the surface pits and scratches are significantly shallower. In particular, it shows the smoothest wear mark under the condition of 0.2 wt% IF-WS_2-300r-5N, which reveals that the lubrication effect of the modified IF-WS_2 and the optimal condition (300r-5N). The above results of the surface morphology differences caused by the modified IF-WS_2 additive are basically consistent with the wear analysis results of the counterbodies (steel plates). Besides, combining with the three-dimensional morphology, surface morphology and the counterbodies (steel plates) wear surface pictures described above, it is not difficult to find the material transfer at the corresponding points, especially under the condition of castor oil lubrication, which verifies from the side that one of the main wear mechanism under the condition of base oil is adhesion wear. In addition, from the cross-section profile data showed in Fig. 15 (aIII)-(fIII), it is clearly that the width and depth of the abrasion marks lubricated by the lubricating oil containing 0.2 wt% modified IF-WS_2 will decrease significantly compared with the base oil under the same conditions, so that to reduce wear volume greatly (the blue dotted line represents the fitting contour of the original steel balls). The results are also consistent with the wear analysis results of the counterbodies (steel plates). What is more important is that from cross-section profile and microscopic morphology diagrams, we could find that compared with that of base oil, the furrows with different depths on the scratches lubricated by the lubricating oil containing 0.2 wt% modified IF-WS_2 are more obvious, which is obviously a change of wear form from the adhesion wear mentioned above to abrasive wear. The following section on the characterization of wear scars demonstrates this opinion.

In conclusion, the above analysis results indicate that the wear analysis results of the steel balls are basically consistent with those of the counterbodies (steel plates) wear analysis, and both can represent the wear analysis results of the tribosystem to a large extent. Therefore, for simplicity, the following section on the characterization of wear scars will be analyzed only from the steel plates and the steel balls' analysis will be neglected.

3.3. Characterizations of wear scar

3.3.1. Surface morphology

The influence of modified IF-WS_2 particles on the microscopic morphology of wear scars is characterized by SEM. The SEM diagrams of the wear scars under various experiment conditions are showed in Fig. 16. As showed in Fig. 16 (a), it can find lots of pits and deep grooves on the wear scar lubricated by base lubricant at the conditions of 250 rpm and 4 N. Combined with the analysis of friction-pairs' wear material transfer in the third chapter, they prove that there are serious adhesive wear between the friction parts when lubricate by base oil.

In contrast, under the same conditions, the worn surface lubricated by castor oil with 0.2 wt% modified IF-WS_2 shown in Fig. 16 (b) displays a smoother surface with almost no pits. However, when it is lubricated by castor oil with 0.5 wt%, pits and furrows reappear, and there are even some rough bumps on the worn surface (Fig. 16 (c)). As for Fig. 16 (d), (e) and (f), they show the worn surface in the case of 50 rpm, 300 rpm and 500 rpm (0.2 wt%, 4 N). It is obvious that the most smooth worn surface, having few scratches, occurs at 300 rpm (Fig. 16 (e)). Although it is possible that the speed has few influence on wear mark roughness, the surfaces of the wear scars at very low or very high speeds (Fig. 16 (d) and (f)) are still rougher than that at 300 rpm. Similar to speed, it shows the smoothest worn surface at 5 N in Fig. 16 (h), although the wear scar width increases slightly with the increase of load, the worn surface of 5 N is smoother than that of 3 N manifested in that there are more pits and deep scratches at 3 N compared with 5 N. However, when the applied load increases to 10 N, many serious pits, furrows and coarse protrusions show up on the wear scar showed in Fig. 16 (i). The extremely rough surface means that there are serious abrasive wear and adhesive wear in the experiment.

All of the above analyses of the worn surface show a conclusion that is under the conditions of 300 rpm and 5 N, the wear scar lubricated by the castor oil with 0.2 wt% modified IF-WS_2 is the smoothest. This

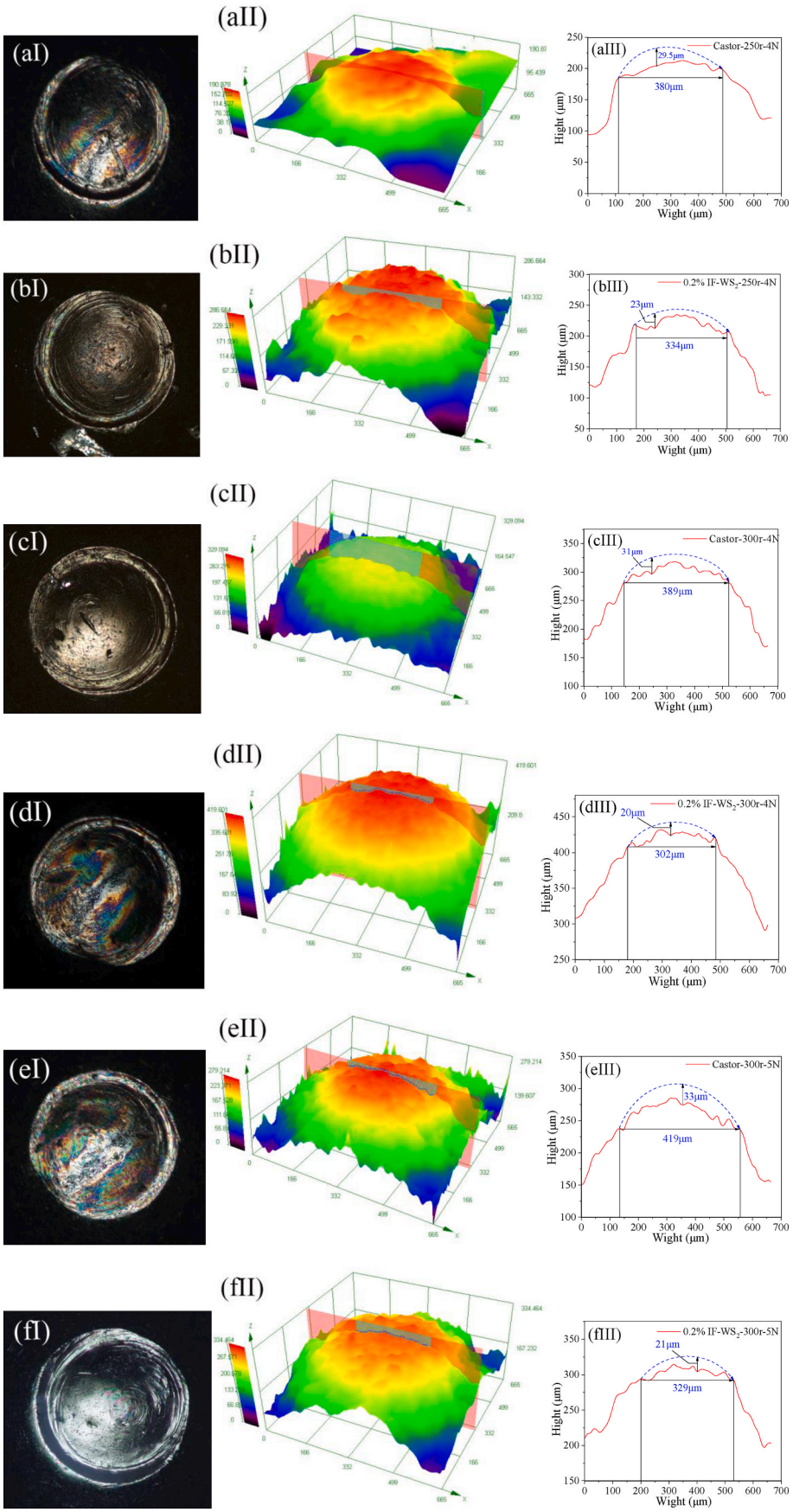

Fig. 15. The microscopic morphology, three-dimensional image and cross-section profile of the wear marks of the steel balls. (aI)-(aIII) castor-250r-4N; (bI)-(bIII) 0.2% IF-WS_2-250r-4N; (cI)-(cIII) castor-300r-4N; (dI)-(dIII) 0.2% IF-WS_2-300r-4N; (eI)-(eIII) castor-300r-5N; (fI)-(fIII) 0.2% IF-WS_2-300r-5N.

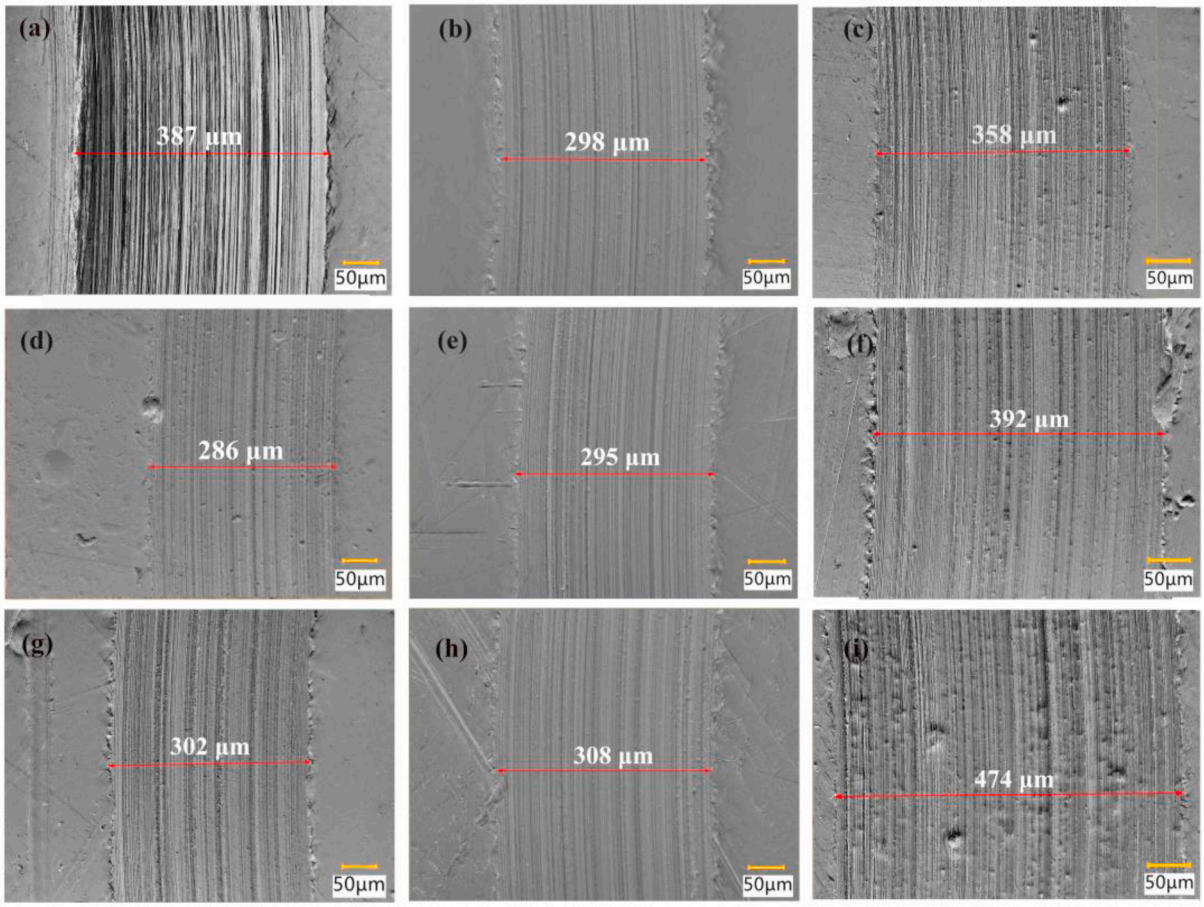

Fig. 16. SEM images of wear surface. (a), (b) and (c) appearances of wear scar at 0 wt%, 0.2 wt%, 0.5 wt% cases (250 rpm, 4 N); (d), (e) and (f) appearances of wear scar at 50 rpm, 300 rpm and 500 rpm cases (0.2 wt%, 4 N); (g), (h) and (i) morphology of worn surface at 3 N, 5 N and 10 N cases (0.2 wt%, 300 rpm).

conclusion is mutually verified with the previous conclusion obtained from the analysis of friction-reduction and anti-wear performance. Moreover, it is not difficult to find that even when lubricating with lubricants containing additives, there will still be furrows under heavy load conditions, and even aggravation of furrows, which should be attributed to the increase in abrasive wear caused by the additives themselves. However, due to the unique friction characteristics of additives, the wear amount caused by abrasive wear is relatively small. This opinion will be verified by Raman and EDS analysis in Sections 3.3.2-3.3.3.

3.3.2. *Raman spectrum of wear scar*

To further investigate the lubrication mechanism, the Raman spectra of the IF-WS_2, 2H-WS_2 samples and the worn surface, shown in Fig. 17, are obtained. We can clearly observe that there are two different strong Raman peaks which are located at 351.0 cm^{-1} and 416.8 cm^{-1}, respectively. The two peaks correspond to E^1_{2g} and A_{1g} modes of first order Raman peaks, and the E^1_{2g} and A_{1g} modes represent the W + S motion on the x-y plane and the two S atoms motion along the z-axis of the cell, respectively [41–43]. As can be seen in Fig. 17 (a) and (b), there is a high similarity between the Raman spectra of the IF-WS_2 and the 2H-WS_2. However, they also exist subtle differences. On the one hand, it is reflected in the wavelength at around 420 cm^{-1}, where the Raman spectrum of IF-WS_2 has a more obvious splitting peak at the wavelength of 420 cm^{-1} compared with 2H-WS_2, which may be caused by second-order Raman effect [17]. On the other hand, in the E^1_{2g} mode, 2H-WS_2 has a wider Raman peak, while in the A_{1g} mode, it has a sharper Raman peak. In Fig. 17 (c), no Raman spectral signal is detected on the wear scar lubricated by castor oil. On the contrary, two strong signals were detected at 350 cm^{-1} and 420 cm^{-1} on the worn surface in the case of 0.2 wt% modified IF-WS_2, which proves that the modified IF-WS_2 nanoparticles indeed enter the sliding interface and subsequently form tribofilm.

Although two Raman peaks signal can be detected in the case of 0.2 wt% modified IF-WS_2 at any conditions, the shape of Raman peaks are inconsistent under different conditions, verified in Fig. 17 (d) and (e). In Fig. 17 (d), comparing with the case of 500 rpm, it is easy to observe that the peak of E^1_{2g} at 355 cm^{-1} is slightly broadened and the peak of A_{1g} at 420 cm^{-1} is slightly sharper under 300 rpm condition, attributing to the increase of 2H-WS_2 content in the worn surface due to the exfoliation and damage under the pressure and shear action [44,45]. When at 500 rpm, a higher rotating speed makes the friction pair enter into steady wear period rapidly, accompanied by the decreasing Hertz contact pressure. Therefore, under lower pressure and shear, the exfoliation and damage on spherical layered IF-WS_2 are slight. In contrast, at 300 rpm, the exfoliation and damage effects are tinily increased, resulting in an increase in 2H-WS_2, this can be mutually verified by the changes of Raman peaks. The Raman images of wear scar under variable load conditions are shown in Fig. 17 (e), similar to the rotate speed conditions, in the case of 10 N, due to the high pressure and shear effect, the outer layers of the IF-WS_2 are exfoliated in large quantities, and some particles are even damaged, leading to a great increase of 2H-WS_2 nanoparticles on the friction surface, so that theE^1_{2g}peak was broadened and the A_{1g} peak sharpened comparing with the case of 5 N.

Besides, two different kinds of Raman spectra, which represent the IF-WS_2 and the 2H-WS_2 respectively, can be detected at any conditions. However IF-WS_2 samples can only detect the IF-WS_2 Raman spectra.

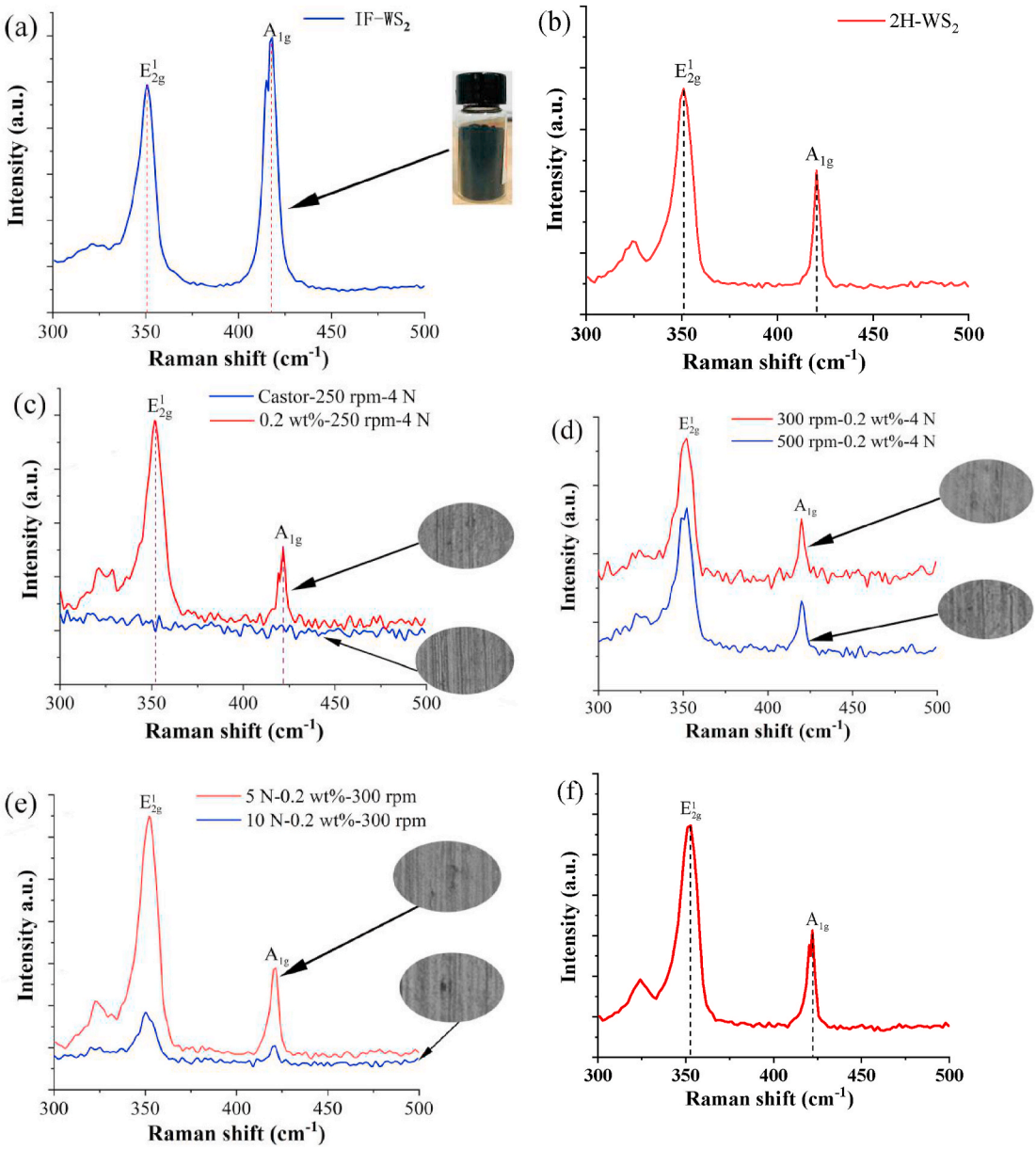

Fig. 17. Raman spectrum images of samples and worn surfaces. (a) IF-WS_2 sample; (b) 2H-WS_2 sample; (c) worn surface in different concentrations; (d) worn surface in different rotate speeds; (e) worn surface in different loads; (f) worn surface in any 0.2 wt% modified IF-WS_2 case.

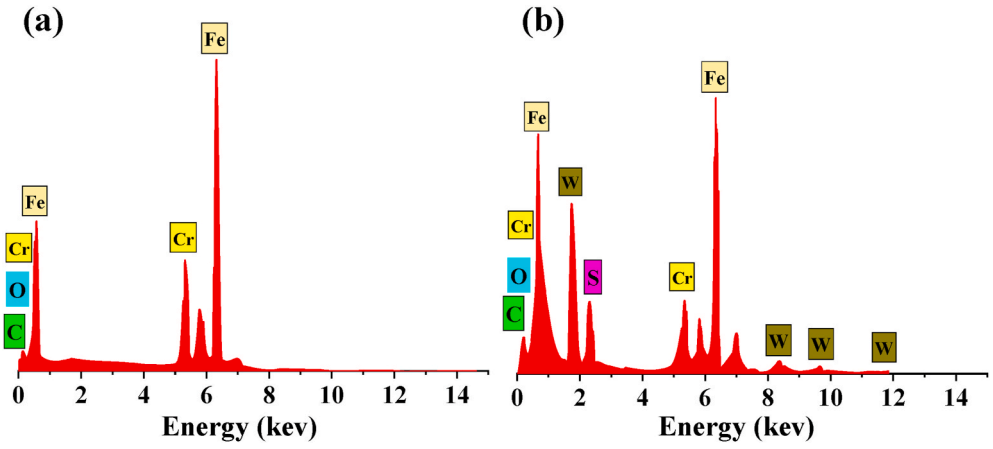

Fig. 18. EDS analysis diagrams of element distribution. (a) Castor oil; (b) 0.2 wt% OA/IF-WS_2.

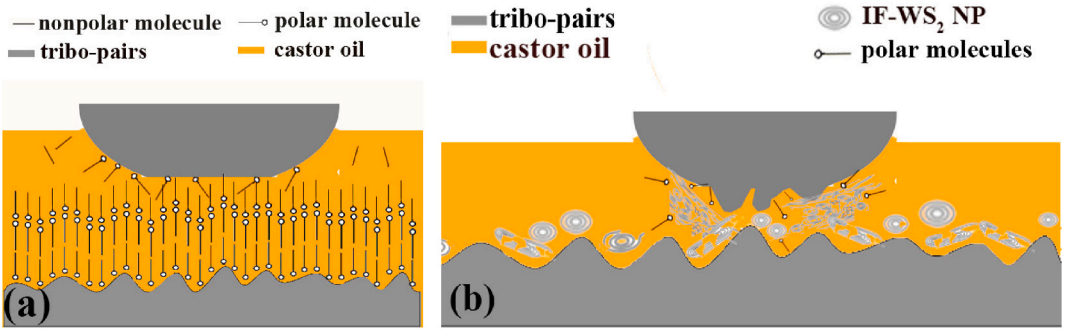

Fig. 19. Simplified images of the lubrication mechanism. (a) pure castor oil; (b) castor oil with 0.2 wt% modified IF-WS_2 at low speed heavy load.

Fig. 17 (f) is a representative IF-WS_2 Raman spectra in different conditions which is different from Fig. 17 (d) and (e). Moreover, the IF-WS_2 nanoparticles have stable structures that are difficult to exfoliate and react under natural conditions. In conclusion, it can verify that the 2H-WS_2 is exfoliated from the IF-WS_2 and then they deposit together with the IF-WS_2 to form a tribofilm.

The above analysis shows that modified IF-WS_2 nanoparticles enter the sliding interface and form deposited tribofilms on the worn surface, resulting in the reduced friction and wear. In addition, due to the wear effect, the particle structure has been changed, reflecting in the shape change of Raman peaks.

3.3.3. *EDS analysis of the wear scars*

Fig. 18 shows the EDS data obtained from the wear marks lubricated by base oil and 0.2 wt% OA/IF-WS_2. It is obvious in Fig. 18 (a) that no sulfur or tungsten elements are detected in the wear areas lubricated by the base oil. In addition to carbon element, other detection elements are the constituent elements of SUS304 steel sheet. As for carbon element, on the one hand, it may be related to the original composition of SUS304 steel. On the other hand, it may be attributed to the carbonization of base oil during friction process.

However, when lubricated by 0.2 wt% OA/IF-WS_2, from Fig. 18 (b), comparing with the base oil, it can be found that sulfur and tungsten elements increased significantly. Moreover, the results of further quantitative analysis of elements indicate that W:S = 1:1.9 which is slightly smaller than the element ratio of IF-WS_2. It may be due to the presence of a small amount of unvulcanized WO_x in the center of IF-WS_2. Therefore, the Raman and EDS characterizations together prove that OA/IF-WS_2 can deposite on the friction surface to form a deposition film, causing reduced friction and wear.

Raman and EDS analysis findings together demonstrate that the modified IF-WS_2 is involved in the friction surface, combined with wear analysis in third part, they prove together that the wear mechanism with additives is mainly abrasive wear causing by additives (modified IF-WS_2).

3.4. *Lubrication mechanism of the modified IF-WS_2*

For the appearance of optimum rotate speed, a possible explanation is that the oil film thickness is thin due to the low rotational speed, making only a little modified IF-WS_2 nanoparticles play its role under such condition. With higher speed applied, the thicker oil film is formed, which makes the lubrication state turn into elastohydrodynamic lubrication, causing the enhanced effect of castor oil. As for the appearance of optimum load, it may attribute to that when the load is low, the nanoparticles lose its function because of the thicker oil film and higher oil viscosity. In contrast, the oil film thickness and viscosity of the lubricant decrease to minimum under high load, thus oil film is extremely easy to break. Therefore, lubrication is mainly performed by the nanoparticles. In summary, only in the case of 300 rpm and 5 N, castor oil can interact perfectly with 0.2 wt% modified IF-WS_2 nanoparticles to provide the best lubrication effect. Therefore, with the increase of speed and load, the average friction coefficient decreases but increase subsequently. Meanwhile, the relative wear widths first increase and then decrease. The action process of IF-WS_2 particle is described detailedly below.

When the steel/steel contact surface is lubricated by pure castor oil (Fig. 19 (a)), since castor oil is mainly composed of polar molecules, these polar molecules will be physically adsorbed on metal surface because of the stronger van der Waals force of the polar molecules, thus forming more regular arrangement of nano tribofilm to promote effective lubrication [46–51]. However, because of the relatively weak van der Waals force existed in the polar molecules, polar molecules have limited adsorption capacity. In fact, the lubrication effect is poor.

In contrast, the lubrication of the mixed lubricant is better at proper rotational low speed and heavy load. The lubrication mechanism can be illustrated by Fig. 19 (b). Due to the effect of pressure and shear, the outer layers of some IF-WS_2 nanoparticles are exfoliated, then the exfoliated layered tungsten disulfide and the original structural IF-WS_2 nanoparticles are deposited together on the friction surface to form a tribofilm. In fact, the above viewpoint has been proved in the characterization of wear marks by Raman spectroscopy, that is the obvious characteristic peak of 2H-WS_2 was detected on the corresponding wear scar. At the same time, polar lubricating oil molecules and the oleic acid attached to the outer layer of inorganic fullerene through chemical bond will also be more uniformly adsorbed near the modified IF-WS_2, making the additives better adsorb on the friction surface to reduce friction and wear. Even if the outer layer is peeling off, it will be rapidly deposited and adsorbed under the action of the modified oleic acid. It prevents the direct contact between friction pairs better, thus improving the lubrication effect.

4. Conclusions

When IF-WS_2 nanoparticles modified by oleic acid are used as the additives of castor oil, the dispersibility of additive particles is improved greatly. Moreover, the tribological properties of the mixed lubricant with IF-WS_2 as additives are significantly improved compared with that of pure castor oil. Especially at 0.2 wt% case, the friction coefficient and worn volume can be reduced by 27.4% and 47.2%, respectively, under 300 rpm and 5 N condition. In addition, the improvement of tribological performance is also manifested in smoother worn surfaces, which can be seen by SEM images. Meanwhile, the reason, validated by Raman spectroscopy and EDS spectroscopy, for the improvement in tribological performance is that IF-WS_2 particles have entered the friction interface, thus preventing the direct contact between the steel ball and plate. Furthermore, because of the layered spherical structure of the IF-WS_2 and the influence of pressure and shear, parts of the outer layer will be exfoliated and attached to the worn surface, thus further decreasing the friction and worn loss. This improvement in tribological properties is due to the unique properties of the additive IF-WS_2 and the polarity of castor oil. Therefore, the green pollution-free mixed lubricant formed by the combination of IF-WS_2 nanoparticles and vegetable lubricating oil

(castor oil) has a great application prospect in the field of tribology.

Author contribution statement

Tiancheng Ouyang: Formulation of overarching research goals and experimental scheme. **Wenwu Lei:** Experimental test, Formal analysis, English writing. **Wentao Tang:** Data collation and material characterization. **Yudong Shen:** Material characterization. **Chunlan Mo:** English revision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

a: radius of the contact circle
E_1: Young's modulus of the steel ball
E_2: Young's modulus of the steel plate
IF-WS_2: Inorganic fullerene tungsten disulfide
IF-MoS_2: inorganic fullerene molybdenum
OA/IF-WS_2: inorganic fullerene tungsten disulfide modified by oleic acid
P: normal load
q_0: the maximum Hertz contact pressure
R_0: radius of the steel ball
WO_3: tungsten trioxide
WO_x: tungsten oxide
2H-WS_2: layered tungsten disulfide
-OH: hydroxyl groupGreek symbols
Ø: diameter
μ_1: Poisson's ratio of the steel ball
μ_2: Poisson's ratio the steel plate
δ: the change value in distance from the central point of the steel sphere to the plane of the steel sheets caused by elastic deformation**Abbreviations**
CNTs: carbon nanotubes
EDS: energy dispersive spectrometer
FTIR: Fourier transform infrared spectrometer
JCPDS: Joint Committee on Powder Diffraction Standards
NP: nanoparticles
OA: oleic acid
PAO: poly alpha olefin
SEM: scanning electron microscope
TEM: transmission electronic microscope
XRD: X-ray diffractometer